



Team Information

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Problem Statement

Compliance to the Republic Act of 11285, along with ISO 50001 and 50002, for energy efficiency and conservation across industrial and commercial sectors [1] requires data extraction and consolidation in energy audit reporting. This often involves manually gathering energy-related data from diverse and unstructured sources and in turn is time-consuming and prone to errors. Errors include examples such as misreading figures, missing important information, or duplicating entries. This makes it difficult to create accurate and timely reports, especially when large amounts of data need to be processed under tight deadlines [2]. Organizations also face challenges in monitoring, and optimizing energy use, which leads to missed opportunities for cost savings, efficiency, sustainability, and regulatory compliance [3].

Proposed Solution

EnergAlze — is a web-based NLP system that streamlines the process of extracting data from spreadsheets. It is powered by a machine learning model trained to process and summarize volumes of energy data to produce summaries and insights. By automating the extraction, organizations can identify actionable insights for optimizing energy use that will lead to cost savings and compliance with regulatory requirements.

Target Market

Designated establishments (companies, organizations, institutions that have 50,001KWH and above annual consumption) and Certified Energy Auditors required to submit audit reports under R.A. 11285 will benefit significantly from using **EnergAlze**, while the Department of Energy will gain from receiving accurate, timely reports to

enhance compliance monitoring and drive energy-saving efforts.

Key Features

By analyzing and summarizing key insights, trends, and actionable points from the extracted data, **EnergAlze** can save significant time for auditors and decision-makers, making it easier for them to focus on actionable insights rather than sifting through raw data. **EnergAlze** will also implement Regression Analysis to predict future energy consumption and identify areas of inefficiencies.

Expected Outcome

- Enhanced efficiency and accuracy of data analysis
- Improved decision making for the stakeholders
- Scalability for handling larger datasets as time goes by
- Streamlining the process of creating audit reports
- A website that generates summaries based on training data and datasets.
- Forecast future energy usage and identify inefficiencies with predictive analytics

NLP applications to be used

Text Generation: Generate complete energy audit reports with recommendations.

Text Summarization: Summarize key trends and inefficiencies from energy data.

References

- [1] R.A. 11285, "An Act Institutionalizing Energy Efficiency And Conservation, Enhancing The Efficient Use Of Energy, And Granting Incentives To Energy Efficiency And Conservation Projects," Official Gazette of the Republic of the Philippines, 2019. Available: <https://www.officialgazette.gov.ph/downloads/2019/04apr/20190412-RA-11285-RRD.pdf>
- [2] "Navigating the Challenges of Data Consolidation for Informed Decision-Making and Maximized ROI," [www.linkedin.com. https://www.linkedin.com/pulse/navigating-challenges-data-consolidation-informed-roi-subhashini/](https://www.linkedin.com/pulse/navigating-challenges-data-consolidation-informed-roi-subhashini/)
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